Strand 3 · Geometry and Measurement

This strand (about 13 of 40 questions) covers **dilations, transformations, angles, volume, surface area, and the Pythagorean theorem.** Keep the **reference sheet** open — most of these use a formula.

Standards covered:

- ○ **8.3A** — Ratio of corresponding sides of similar shapes is proportional. ○ **8.3B** — Compare a shape and its dilation on a plane. ★ **8.3C** — Algebraic rule for a scale factor (origin center).
- ○ **8.6A** — Volume formula V = Bh of a cylinder. ○ **8.6C** — Explain the Pythagorean theorem with models.
- ★ **8.7A** — Volume of cylinders, cones, spheres. ★ **8.7B** — Surface area of prisms and cylinders. ★ **8.7C** — Pythagorean theorem and its converse. ○ **8.7D** — Distance between two points.
- ○ **8.8D** — Angle sums, exterior angles, transversal angles, AA similarity.
- ○ **8.10A** — Orientation/congruence of transformations. ○ **8.10B** — Which preserve congruence. ★ **8.10C** — Rules for translations, reflections, rotations (90/180/270/360). ○ **8.10D** — Effect of a dilation on length and area.

Part A · Dilations and Similarity (8.3A ○, 8.3B ○, 8.3C ★, 8.10D ○)

A **dilation** resizes a figure by a **scale factor k** from a center. With the **origin** as center: **(x, y) → (kx, ky).**

- **8.3A / 8.3B — Similar shapes.** A figure and its dilation are **similar**: corresponding sides are proportional (×k) and angles stay the **same.**
- **8.10D — Effect on measurements.** Lengths multiply by **k**; **area** multiplies by $\mathbf{k^2}$**.**

Worked example (8.3C). Vertex (2, 3) dilated by factor 3 about the origin → (3·2, 3·3) = **(6, 9).**

Watch Out: k > 1 enlarges; 0 < k < 1 shrinks. Area changes by the **square** of k — factor 2 makes area **4×**.

Try It. A square of side 5 is dilated by k = 2 — new side? *(10.)*

Part B · Transformations on the Coordinate Plane (8.10A/B ○, 8.10C ★)

Transformation	Algebraic rule	Changes size?
Translation (slide)	(x, y) → (x + a, y + b)	No — congruent
Reflection over x-axis	(x, y) → (x, −y)	No — congruent
Reflection over y-axis	(x, y) → (−x, y)	No — congruent
Rotation 180° about origin	(x, y) → (−x, −y)	No — congruent
Rotation 90° counterclockwise	(x, y) → (−y, x)	No — congruent

Dilation (factor k)	(x, y) → (kx, ky)	**Yes** — similar, not congruent

8.10A / 8.10B — Congruence. Translations, reflections, and rotations are **rigid** — the image is **congruent.** A dilation (k ≠ 1) changes size, so it is **not** congruent, only **similar.**

Worked example (8.10C). Reflect (4, −2) over the x-axis → **(4, 2).**

Watch Out: Reflecting over the **x-axis** flips the **y**-sign; over the **y-axis** flips the **x**-sign.

Try It. Rotate (3, 5) by 180° about the origin. *((−3, −5).)*

Part C · Angle Relationships (8.8D ○)

- **Triangle angle sum:** interior angles add to **180°.**
- **Exterior angle** = sum of the two remote interior angles.
- **Parallel lines + transversal:** corresponding angles equal; alternate interior angles equal.
- **Angle-Angle (AA) similarity:** two equal angles → triangles are **similar.**

Worked example. Angles 50° and 60° → third = 180 − 50 − 60 = **70°.**

Try It. Angles 90° and 35° — find the third. *(55°.)*

Part D · Volume (8.6A ○, 8.7A ★)

- **Cylinder:** V = Bh = $\pi r^2 h$. **Cone:** V = $\frac{1}{3}\pi r^2 h$ (one-third of the cylinder). **Sphere:** V = $\frac{4}{3}\pi r^3$.

Worked example (8.7A). Cylinder radius 3, height 10: V = $\pi(3^2)(10)$ = **$90\pi \approx 282.6$** cubic units.

Watch Out: The cone is ⅓ of the matching cylinder — don't forget the one-third.

Try It. Cone radius 3, height 10 — volume? *(⅓·90π = 30π ≈ 94.2.)*

Part E · Surface Area (8.7B ★)

- **Prism — total:** S = Ph + 2B. **Cylinder — total:** S = $2\pi rh + 2\pi r^2$ (side plus two circles).

Worked example (8.7B). Cylinder radius 2, height 5: S = $2\pi(2)(5) + 2\pi(2^2) = 20\pi + 8\pi$ = **$28\pi \approx 87.9$** square units.

Strategy: "Lateral" = the **side** only; "total" **adds the bases.** Read which one is asked.

Try It. Lateral surface area of that cylinder? *($2\pi rh = 20\pi \approx 62.8$.)*

Part F · The Pythagorean Theorem (8.6C ○, 8.7C ★, 8.7D ○)

For a **right triangle** with legs a and b and hypotenuse c: $\mathbf{a^2 + b^2 = c^2}$.

- **8.7C** — Find a missing side, or use the **converse** to test whether a triangle is right.

- **8.7D** — Distance between two points: horizontal and vertical gaps are the legs; the distance is the hypotenuse.

Worked example (8.7C). Legs 6 and 8: $c^2 = 36 + 64 = 100 \rightarrow c =$ **10.** **(8.7D)** Distance (1, 2) to (4, 6): legs 3 and 4 $\rightarrow \sqrt{(9 + 16)} =$ **5.**

Watch Out: The hypotenuse **c** is across from the right angle and is the **longest** side.

Try It. Is 5, 12, 13 a right triangle? *(Yes — 25 + 144 = 169 = 13^2.)*

Strand 3 Checklist

- ☐ 8.3A–C, 8.10D Dilate figures; use scale factor; length ×k and area ×k^2
- ☐ 8.10A–C Apply translation, reflection, rotation rules; know what stays congruent
- ☐ 8.8D Use angle sums, exterior angles, transversals, AA similarity
- ☐ 8.6A, 8.7A Volume of cylinders, cones, spheres
- ☐ 8.7B Lateral and total surface area of prisms and cylinders
- ☐ 8.6C, 8.7C, 8.7D Pythagorean theorem, converse, and distance